3 **(a)** **(i)**	$\omega = 2\pi / T$		
	$= 2\pi / 0.69$	C1	
	$= 9.1 \text{ rad s}^{-1}$	A1	[2]
	(allow use of $f = 1.5$ Hz to give $\omega = 9.4 \text{ rad s}^{-1}$)		
(ii) **1.**	$x = 2.1 \cos 9.1t$		
	2.1 and 9.1 numerical values	B1	
	use of cos	B1	[2]
2.	$v_0 = 2.1 \times 10^{-2} \times 9.1$ (allow ecf of value of x_0 from **(ii)1.**)		
	$= 0.19 \text{ m s}^{-1}$	B1	
	$v = v_0 \sin 9.1t$ (allow $\cos 9.1t$ if sin used in **(ii)1.**)	B1	[2]
(b)	energy = either $\frac{1}{2} m v_0^2$ or $\frac{1}{2} m \omega^2 x_0^2$		
	= either $\frac{1}{2} \times 0.078 \times 0.19^2$ or $\frac{1}{2} \times 0.078 \times 9.1^2 \times (2.1 \times 10^{-2})^2$	C1	
	$= 1.4 \times 10^{-3}$ J	A1	[2]